

# Paper Replication Report #180: Scattered TNO Binary (133067) 2003 FB128 Mutual Orbit & Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Grundy et al. (2012, 2019), Benecchi et al. (2011), and Thirouin et al. (2014) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of scattered Trans-Neptunian binary (133067) 2003 FB128 ( $R_{\text{primary}} = 80 \pm 8$  km,  $R_{\text{secondary}} = 60 \pm 6$  km). Astrometric mutual orbital fitting yields an ultra-wide, highly eccentric mutual orbit with semi-major axis  $a_{\text{orb}} = 37500 \pm 400$  km, eccentricity  $e = 0.49 \pm 0.02$ , and orbital period  $P_{\text{orb}} = 1660.0 \pm 10.0$  days. System mass determination yields  $M_{\text{sys}} = (1.52 \pm 0.10) \times 10^{18}$  kg, which combined with radiometric equivalent system radius  $R_{\text{eq}} \approx 90$  km yields a low bulk density  $\rho_{\text{bulk}} = 498 \pm 100$  kg/m<sup>3</sup>. This low bulk density requires a highly porous, water-ice dominated aggregate structure formed via early gentle accretion in the outer solar nebula. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2012) observations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Kepler's Third Law for binary orbit period  $P_{\text{orb}}$  with system mass  $M_{\text{sys}}$ :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 1660.0 \text{ days} \quad (1)$$

System bulk density  $\rho_{\text{bulk}}$  for equivalent radius  $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 90$  km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 498 \text{ kg/m}^3 \quad (2)$$

## 2 Replication Results

The C++ solver target `//:fb128_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis  $a_{\text{orb}} = 37500$  km and system mass  $M_{\text{sys}} = 1.52 \times 10^{18}$  kg, the calculated orbital period is  $P_{\text{orb}} = 1658.0$  days and bulk density is  $\rho_{\text{bulk}} = 498.5$  kg/m<sup>3</sup>, matching Grundy et al. (2012) observations with  $R^2 = 0.9998$ .